Dr Yidan Xue

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Research Interests

In-silico trials, medical devices, computational medicine, regulatory science, scientific computing, mathematical modelling, machine learning, microcirculation, oxygen transport, Stokes flow.

Education

University of Oxford

- DPhil in Engineering Science, Jan 2023. (Supervisor: Stephen Payne)
- Thesis: Modelling oxygen transport and tissue damage in the human brain

The University of Edinburgh

- BEng (Hons) in Mechanical Engineering, Jul 2019.

Xiamen University

- Attended Sep 2015 – Jun 2017.

Employment

The University of Manchester

- BHF/UK CEiRSI Transition Fellow, Department of Computer Science, Apr 2026 – Present.
- Research Associate, School of Health Sciences, Aug 2024 – Mar 2026.

Cardiff University

- Research Associate, School of Mathematics, Jan 2024 – Jul 2024.

University of Oxford

- EPSRC Postdoctoral Research Associate, Mathematical Institute, Oct 2022 – Dec 2023.
- Retained Lecturer in Mathematics, Jesus College, Jan 2023 – Sep 2023.

Chinese Academy of Sciences

- Research Intern, Institute of Mechanics, Jul 2018 – Sep 2018.

The University of Edinburgh

- Research Intern, School of Engineering, May 2018 – Jul 2018.

Research Funding and Fellowships

(2026) **BHF Manchester Centre of Research Excellence Transition Fellowship**, *The British Heart Foundation (BHF) Manchester Centre of Research Excellence (CRE) & UK Centre of Excellence on In Silico Regulatory Science and Innovation (UK CEiRSI)*. Digital twins to unravel transcatheter heart valves mechanics in bicuspid patients. Two-year fellowship.
(2024) **UK RS&IN Implementation Phase: Human Health (CERSIs)**, *Innovate UK & MRC*, UK CEiRSI - The UK's Centre of Excellence on In-silico Regulatory Science and Innovation - Pilot Phase, Technical Committee Member & Academic Rapporteur of Pilot 4. PI: Alejandro Frangi.

(2022) **EPSRC Postdoctoral Research Associate**, *University of Oxford MPLS Division & EPSRC*. Modelling microthrombi distribution at vessel bifurcations. One-year postdoctoral position awarded to up to 5 DPhil graduates from MPLS Division.
(2018) **Summer Research Scholarship**, *School of Engineering, The University of Edinburgh*. Flow simulation of fuel injection in internal combustion engines. Two-month research project.

Publications

*Author notes: # co-first author; ‡ co-senior author; * corresponding author.*

Journal Articles:

1. **Xue Y**, Sarrami-Foroushani A, Kreuzer S, McVeigh C, Grumbridge M, Bombien R, McLaren A, Pandey PK, Ademiloye AS, Tyler P, MacRaild M, Zakeri A, Revell A, Banda G, Hill D, Nithiarasu P, Manfrin A, De Cunha Maluf-Burgman M, Frangi AF. Risk-informed framework for *in silico* regulatory evaluation of medical devices. *Device*. 2026:101303. **[Press Release]**.
2. Lin F, Zakeri A, **Xue Y**, MacRaild M, Dou H, Zhou Z, Zou Z, Sarrami-Foroushani A, Duan J, Frangi AF. From Pixels to Polygons: A Survey of Deep Learning Approaches for Medical Image-to-Mesh Reconstruction. *Medical Image Analysis*. 2026;113:104153.
3. Jabi W, **Xue Y**, Woolley TE, Kaouri K. 3D Topological Modeling and Multi-Agent Movement Simulation for Viral Infection Risk Analysis. *Architectural Science Review*. 2025.
4. Mao Y, Liu Y, Zhai M, Jin P, Li W, Dong X, Chen F, Wang X, Wang Y, Zhang G, Li H, Yang Y, Zhang H, Liu J, Guo Y, Wu Y, **Xue Y**, Zhang J, Frangi A, Yang J. Precision TAVR Quantification—AI-accelerated TAVR Planning Reduces Assessment Time by 80% in Bicuspid Aortic Stenosis. *European Heart Journal – Imaging Methods and Practice*. 2025;3(4):qyaf153.
5. Payne SJ, **Xue Y**, Kuo J-F, El-Bouri WK. Transit time mean and variance are markers of vascular network structure, wall shear stress distribution and oxygen extraction fraction. *Biomechanics and Modeling in Mechanobiology*. 2025;24:1155–1167.
6. **Xue Y**. Computing Stokes flows in periodic channels via rational approximation. *Proceedings of the Royal Society A*. 2025;481:20240676.
7. **Xue Y**, Payne SJ, Waters SL. Stokes flows in a two-dimensional bifurcation. *Royal Society Open Science*. 2025;12:241392.
8. **Xue Y**, Jabi W, Woolley TE, Kaouri K. Modelling indoor airborne transmission combining architectural design and people movement using the VIRIS simulator and web app. *Scientific Reports*. 2024;14:28220.
9. **Xue Y**, Waters SL, Trefethen LN. Computation of two-dimensional Stokes flows via lightning and AAA rational approximation. *SIAM Journal on Scientific Computing*. 2024;46(2):A1214–A1234. **[SIAM Reproducibility Badge]**.
10. **Xue Y**[#], Georgakopoulou T[#], van der Wijk A-E, Józsa TI, van Bavel E, Payne SJ. Quantification of hypoxic regions distant from occlusions in cerebral penetrating arteriole trees. *PLOS Computational Biology*. 2022;18(8):e1010166.
11. Miller C, Padmos RM, van der Kolk M, Józsa TI, Samuels N, **Xue Y**, Payne SJ, Hoekstra AG. In silico trials for treatment of acute ischemic stroke: design and implementation. *Computers in Biology and Medicine*. 2021;137:104802.
12. **Xue Y**, El-Bouri WK, Józsa TI, Payne SJ. Modelling the effects of cerebral microthrombi on tissue oxygenation and cell death. *Journal of Biomechanics*. 2021;127:110705. **[Special Issue on Thrombus Mechanics]**.

13. **Xue Y**, Hellmuth R, Shin D-H. Formation of vortices in idealised branching vessels: a CFD benchmark study. *Cardiovascular Engineering and Technology*. 2020;11(5):544–559. **[Cover Image]**.

Submitted Articles/Preprints:

14. Padmos RM, Józsa TI, **Xue Y**, Payne SJ, Hoekstra AG. A Multi-Scale Model for Perfusion-Based Infarct Estimation in Acute Ischaemic Stroke Patients. 2023. submitted.
15. Miller C, Padmos R, Konduri P, Józsa TI, **Xue Y**, Arrarte Terreros N, van der Kolk M, Payne SJ, Marquering H, Majoie C, Hoekstra A. In silico trials of acute ischemic stroke: predicting the total potential for improvement to patient functional outcomes. 2025. submitted. Also available on arXiv: `https://doi.org/10.48550/arXiv.2511.02088`.
16. Zhou Z, Zakeri A, Dou H, **Xue Y**, MacRaild M, Huang J, Lin F, Sarrami-Foroushani A, Duan J, Frangi AF. Synthetic Anatomy: Deep Learning Models for Virtual Population Generation—A Review. 2025. submitted. Also available on medRxiv: `https://doi.org/10.1101/2025.10.28.25338782`.
17. Mao Y, Wei H, Qiao Y, Liu Y, Lu F, Wu Y, **Xue Y**, Zakeri A, Sarrami-Foroushani A, Yang Y, Song L, Li Y, Frangi A, Yang J. Automatic Quantification of Aortic Valve Fibrocalcification for Clinical Phenotyping and Prognostic Stratification. 2026. submitted.
18. Dou H[#], **Xue Y**[#], Lin F, Danesh M, Keshavarzi F, Flynn D, Sarrami Foroushani A, Zakeri A, Frangi AF. An automatic workflow for creating digital heart twins at scale for in silico trials of cardiac implants. 2026. submitted.
19. Mao Y[#], Wei H[#], **Xue Y**[#], Liu Y, Wang Z, Frangi AF, Yang J. Digital Twins in Cardiovascular Medicine: Applications, Challenges, and Future Perspectives in the Era of Artificial Intelligence. 2026. submitted.

DPhil Thesis:

20. **Xue Y**. Modelling oxygen transport and tissue damage in the human brain. 2022. DPhil thesis.

Invited Talks

- *From mechanistic modelling to credible in-silico trials: Lessons from cardiovascular medical devices*, From Target Identification to Clinical Trials — Use Case Deep Dive & Topics for 2027, Pistoia Alliance Life Science AI Exchange (LSAIE) Webinar, online, Sep 2026
- *Predicting conduction disturbances after TAVI using patient-specific computational modelling*, BHF Manchester CRE Symposium 2026, Manchester, UK, Aug 2026
- *Towards StressMAP: A TAVI deployment simulation workflow for stress-based surrogate modelling of pacemaker dependency*, The 10th Biennial Heart Valve Biology & Tissue Engineering Meeting, The Royal Society, London, UK, Sep 2025
- *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Physical and Applied Mathematics Seminar, The University of Manchester, UK, Nov 2024
- *Modelling physiological flows and transport at low Reynolds numbers*, CIMIM Seminar (inaugural talk), The University of Manchester, UK, Oct 2024
- *A state-of-the-art epidemic simulator and web app for viral transmission in indoor spaces*, SIAM Conference on the Life Sciences, Portland, Oregon, US, Jun 2024
- *Computation of two-dimensional Stokes flows via lightning and AAA rational approximation*, Computational and Applied Math Seminar, Peking University, China, May 2024
- *Computation of physiological flows and transport at low Reynolds numbers*, Applied and Computational Mathematics Seminar, Cardiff University, UK, Feb 2024
- *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Numerical Analysis

Group Internal Seminar, University of Oxford, UK, May 2023
- *Modelling oxygen transport in the human cerebral microvasculature*, British Applied Mathematics Colloquium, Bristol, UK, Apr 2023

Selected Contributed Talks

- *An automatic workflow for in-silico clinical trials of TAVI devices: From images to deployment and stress analysis*, 2025 MDIC Symposium on Computational Modeling & Simulation, Hyattsville, Maryland, US, Nov 2025
- *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Numerical Analysis in the 21st Century in honour of Nick Trefethen's retirement from Oxford, Oxford, UK, Aug 2023
- *Modelling human cerebral tissue damage caused by acute ischaemic stroke*, 9th World Congress of Biomechanics (WCB), Taipei (online), Jul 2022

Teaching

University of Oxford, Mathematical Institute/Department of Engineering Science
One contact hour requires at least two hours of preparation and marking at Oxford.

Tutor, A1 Differential Equations 1, Oriel College, class size: 1–2, contact hours: 8	Fall 2023
Tutor, A7 Numerical Analysis, Jesus College, class size: 1–2, contact hours: 15	Spring 2023
Tutor, C5.6 Applied Complex Variables, MI, class size: 10–12, contact hours: 16	Spring 2023
Lead Tutor, B17 Biomechanics, EngSci, class size: 3–4, contact hours: 13	Spring 2022

Mentoring

The University of Manchester
MSc Health Data Science 2026
Siru Zhou, *Prediction of Post-treatment Haemodynamic Performance in Aortic Stenosis Patients after TAVI*, co-supervised with Ali Sarrami-Foroushani.
Peixuan Guo, *Prediction of Conduction Abnormalities and Pacemaker Requirement After TAVI Using Clinical and Imaging-Derived Measurements*, co-supervised with Ali Sarrami-Foroushani.
Chen Zheng, *Predicting Six-Month NYHA Non-Improvement After TAVI: Routine Preoperative Assessment and the Incremental Value of Computed Tomography Measurements*, co-supervised with Ali Sarrami-Foroushani.
University of Oxford
MSc Mathematical Modelling and Scientific Computing 2023
Benjamin Nicholls-Mindlin, *Rational Stokes Methods for Tissue Engineering Applications*, co-supervised with Sarah Waters and Helen Byrne.

Awards

(2019) **IMechE Best Student Prize**, The University of Edinburgh.
(2018) **3rd Year Class Medal for Mechanical Engineering**, The University of Edinburgh.
(2018) **Edinburgh Award**, The University of Edinburgh.
(2017/2018) **2+2 Student Scholarships**, The University of Edinburgh.
(2016) **1st Prize Scholarship for Academic Excellence**, Xiamen University.

Professional, Standards and Advisory Roles

ISO/TC 150 Implants for surgery, *Committee Member* 2026 – Present
ISO/TC 150/SC 2/WG 1 Cardiac valves
BSI CH/150 Implants for surgery, *Committee Member* 2026 – Present
BSI CH/150/2 Cardiovascular implants
UK CEiRSI, *Technical Committee Member* 2025 – Present
The Welsh Government, *Policy Modelling Group Member* 2024 – 2025

Scientific Organisation and Leadership

CMBE2027, Credible In Silico Modelling for Regulatory Science and Medical Device Evaluation (mini-symposium), *Organiser (with Adesola Ademiloye, Philippe Favre, Martha De Cunha Maluf-Burgman and Janaki Raman Rangarajan)*, Groningen, The Netherlands, Jun 2027

In Silico Evidence Training for Frontline Regulatory Reviewers, two-day workshop convened at the invitation of IMDRF and coordinated by UK CEiRSI, the Avicenna Alliance and VPH Society, *Co-organiser, Faculty Member and Presenter*, London, UK, Oct 2026

Making Sense of In Silico in Regulatory Affairs, IMDRF regulatory training webinars delivered by UK CEiRSI and the Avicenna Alliance, *Co-organiser and Presenter*, online, Jul 2026

WCCM-ECCOMAS 2026, Advanced Physics-based and Data-driven Modeling for Biomedical Digital Twin (mini-symposium), *Organiser (with Dongwei Ye and Elena Zappon)*, Munich, Germany, Jul 2026

UK CEiRSI In Silico Regulatory Airlock Pilot 4, In-Silico Trials of TAVI Devices: Blood Flow Models Predicting Aortic Stresses and Valve Leakage, *Lead and Academic Rapporteur (pilot meetings); Organiser and Presenter (final webinar), with Martha De Cunha Maluf-Burgman and Ali Sarrami-Foroushani*, Manchester (online), UK, 2025

The 6th China-UK Forum of Young Scholars in Manchester "AI Applications Across Disciplines", *Invited Guest and Panel Judge*, Manchester (online), UK, Jun 2025

Virtual Imaging Trials in Medicine 2025, Innovative Computational Techniques in Medical Imaging, *Session Chair*, Manchester, UK, Jun 2025

Pint of Science (Manchester), *Organiser (with 4 others)*, *Tech Me Out* events, 12 speakers, 160+ attendees, Manchester, UK, May 2025

Numerical Analysis in the 21st Century Conference, in honour of Nick Trefethen's retirement from Oxford, Numerical Methods for Differential Equations, *Session Chair*, Oxford, UK, Aug 2023

Institutional Service

The University of Manchester:
MSc Health Data Science, *Dissertation Marker* 2026
Christabel Pankhurst Institute, *Fire Marshal* 2025 – Present
Recruitment Interview Panel, *Member* 2024, 2026

Cardiff University:
Vendor Selection Panel, *Member* 2024

University of Oxford:
Undergraduate Admissions Panel (Maths & Stats at Oriel College), *Member* 2023
Disability Advisory Service, *Non-Medical Support Worker* 2019 – 2020

Peer Review

Applied & Computational Mathematics: Computational Mechanics, IMA Journal of Numerical Analysis, International Journal of Computer Mathematics, Journal of Computational Physics, Journal of Fluid Mechanics, PLOS Computational Biology, The ANZIAM Journal

Engineering: Biotechnology and Bioengineering, Building Simulation, Cardiovascular Engineering and Technology

Computer Science & AI: IEEE Transactions on Pattern Analysis and Machine Intelligence, Journal of Open Source Software

Interdisciplinary: Journal of the Royal Society Interface

Memberships

EMS Topical Activity Group Mathematical Modelling in Life Sciences, *Member*	2026 – Present
BSI Young Professionals Network, *Member*	2025 – Present
Society for Industrial and Applied Mathematics (SIAM), *Member*	2024 – Present
VPH Institute, *Member*	2020 – 2021, 2024 – Present
European Society of Biomechanics, *Member*	2023 – 2024

Selected Press

The Biotech Times (13 October 2026). Manchester-led team maps path for virtual patients in device regulation.

scinexx (6 October 2026). Virtuelle Patienten sollen Medizinprodukte sicherer machen [German].

Ad-hoc-news / Börse Global (6 October 2026). Medizinprodukte: Neuer Prüfrahmen soll Computersimulationen für Zulassungen bewerten [German].

Medical Xpress (5 October 2026). Virtual patients could help flag medical device risks missed by conventional studies.

The University of Manchester (5 October 2026). New framework aims to make digital evidence credible for medical device regulation [Institutional press release].

SIAM News (13 June 2024). Epidemic Simulator and Web App Models Viral Transmission in Indoor Spaces.